int (all types)
short
char

Range of the data type

Minimum and maximum How long

Formula

$2^{n-1} - 1$

n = size of the data type in bits

int = 4 bytes = 32 bits

$\therefore 2^{n-1}$

$= 2^{32-1}$

$= 2^{31}$

$= -2,147,483,648$ to $2,147,483,647$

Range

int

- signed (-ve & +ve)
- unsigned (only +ve)

Short int = 2 bytes = 16 bits

unsigned 0 to 2^{15}

0 to $(32768) + 2$

= 0 to 65,535

(minus -1)

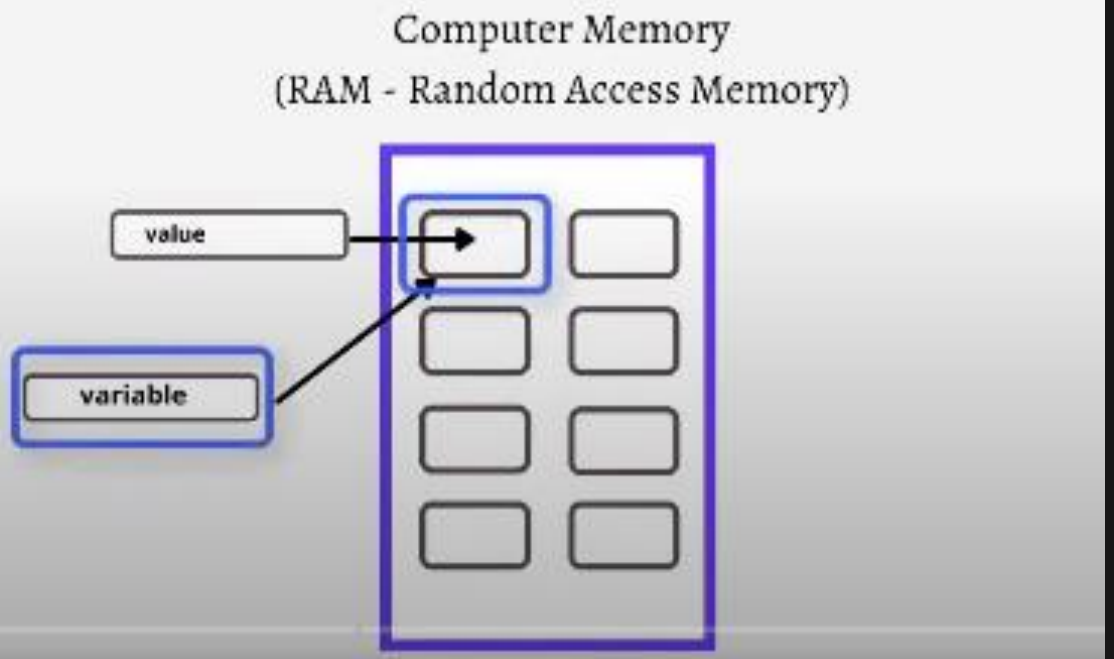
unsigned

0 to $(2,147,483,648) * 2$

= 0 to 4,294,967,295

Variable

- Variable is a name that use to allocate computer memory to store a value
- We can access the stored value using of the variable name
- The allocated memory size depends on **Data Type** of the variable



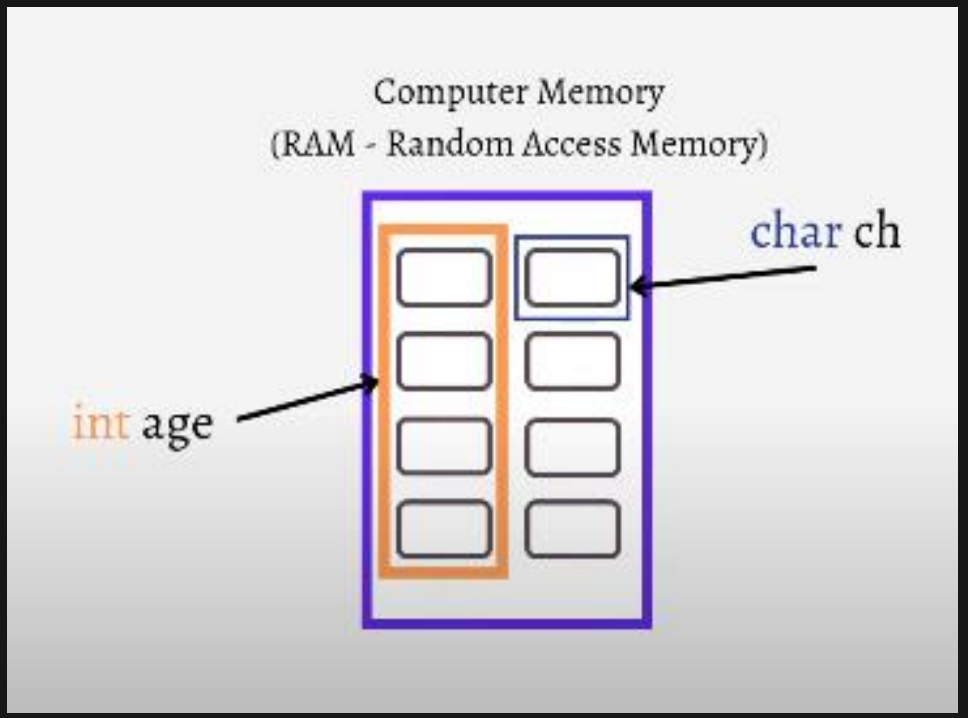
Variable Declaration

Variable Declaration

<data_type> <varaible_name>;

Example:

```
int age;
char ch;
```



Rules for defining a Variable

- A variable name can only have alphanumeric characters (a-z , A-Z , 0-9) and underscore(_). Doesn't allow any other special characters like @, $ etc.

E.g. int first%name;
int num& ;

int first_name;
int age;
int _temp;

- The first character of a variable name can only contain alphabet (a-z, A-Z) or underscore (_). We can't use any digit as a first character

E.g. 2num

int num2

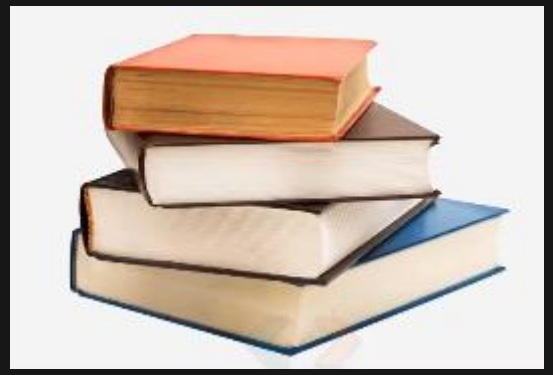
- C Keywords are not allowed to use as a variable name
- C is case-sensitive programing language i.e. Age and age are two different variable in C.

E.g. int int  int INT 

- Blank spaces are not allowed in a variable name.

E.g. int first name;  int first_name; 

DATA TYPE



BOOK

Id	Name	Quantity	Price	Format
1001	C Programming	100	$5.20	H
1002	Java Programming	105	$34.50	P
1003	C++ Programming	20	$20.20	H
1004	Python for Beginners	100	$36.50	P

H = Hardcover Book

P = Paperback Book

What is Data Type

- Data Type defines the **type of data** being used in the program
- According to the Data Type, the **memory allocated** in the computer memory

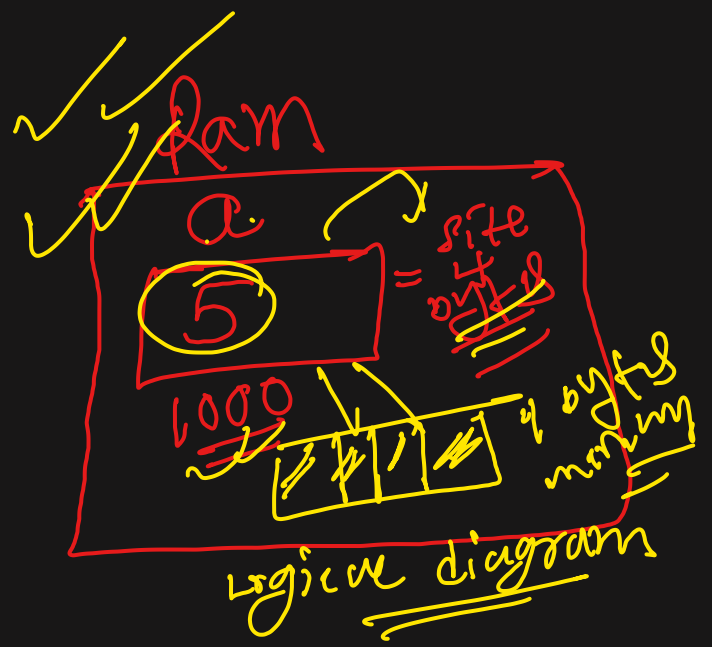
Example

```
int bookId = 1001;
char name[] = "C Programming";
int quantity = 100;
float price = 25.20;
char format = 'H';
```

Book

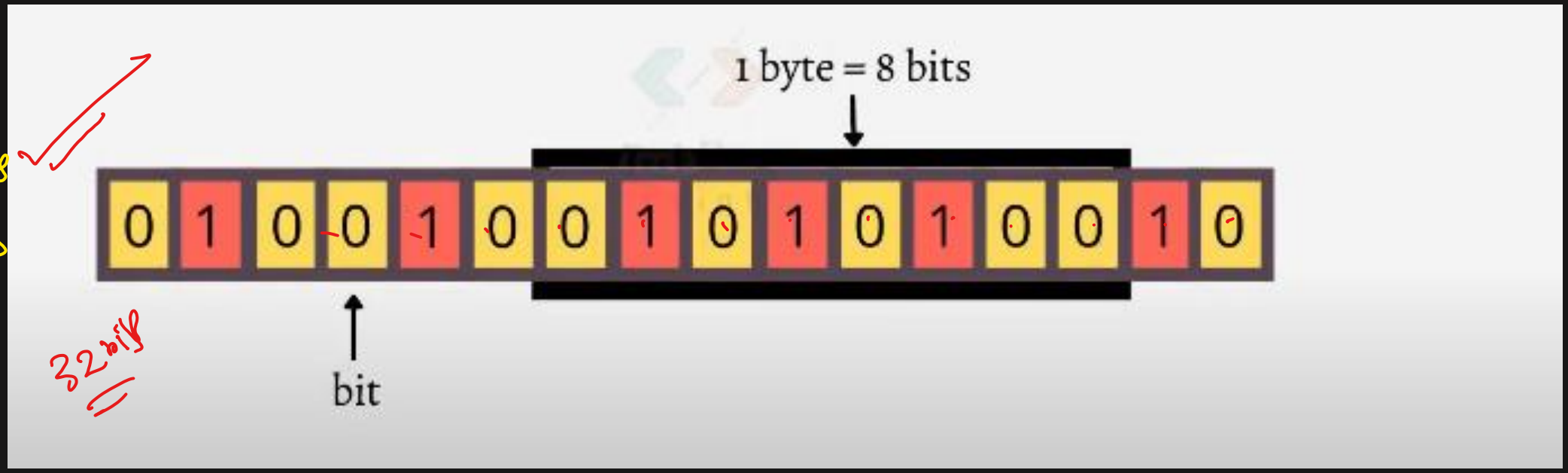
Group of Data Types

1. Primary Data Types
   - int, char, float, double, void
2. Derived Data Types
   - array, structure, union, pointer
3. User-Define Data Types
   - enum, typedef



How data is stored in computer memory

Computer stores information in the form of bytes (1s and 0s) in all its memory.

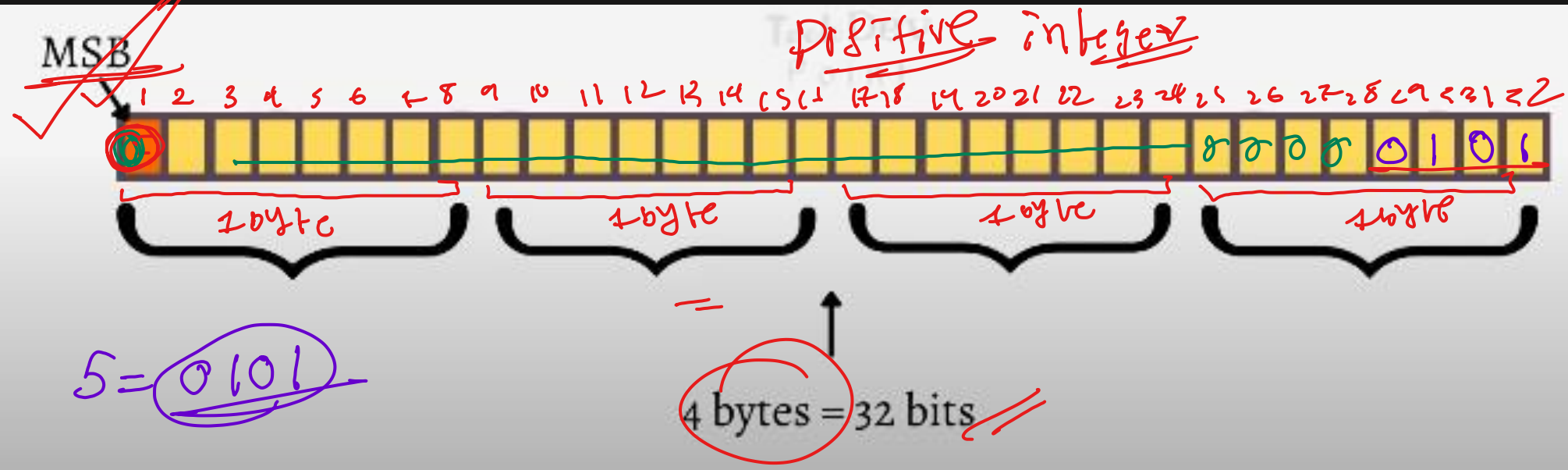


How integers are stored in memory

MSB (most significant bit) = indicate whether the number is positive or negative.

MSB = 0 , for positive numbers

MSB = 1, for negative numbers

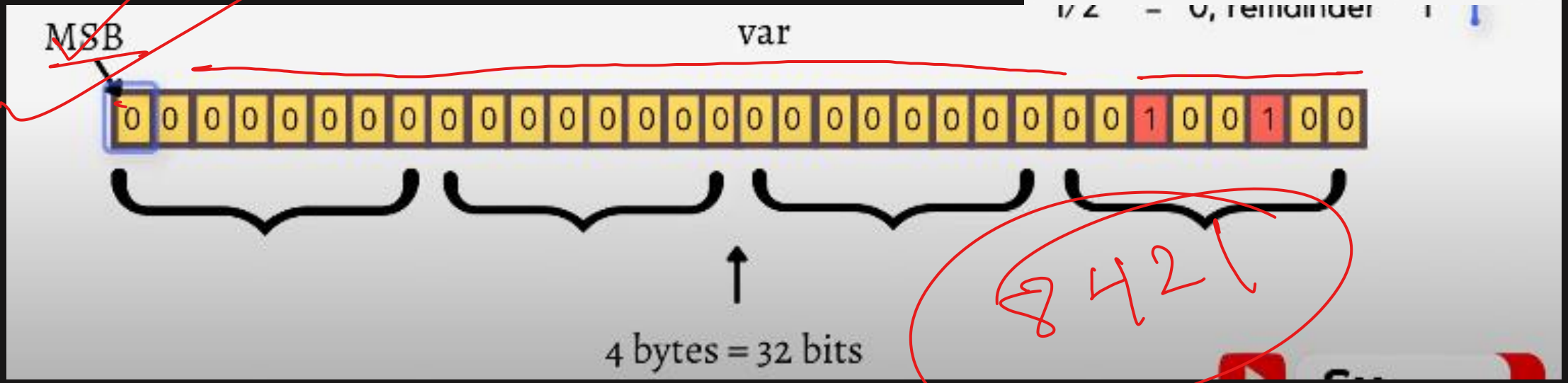


How positive integers are stored in memory

int var = 36;

Binary equivalent of 36 is $(100100)_2$

36/2	= 18,	remainder	0
18/2	= 9,	remainder	0
9/2	= 4,	remainder	1
4/2	= 2,	remainder	0
2/2	= 1,	remainder	0
1/2	= 0,	remainder	1



How negative integers are stored in memory

negative integers are stored as two's complement

Steps to find two's complement:

Step 1: Covert decimal to binary

Step 2: **1's complement** - inverting binary bits (i.e. switching 0 to 1 and 1 to 0)

Step 3: **2's complement** - just add 1 to 1's complement

Binary addition

$$\begin{array}{r} 0 \\ +\,0 \\ \hline 0 \end{array} \qquad \begin{array}{r} 0 \\ +\,1 \\ \hline 1 \end{array} \qquad \begin{array}{r} 1 \\ +\,0 \\ \hline 1 \end{array} \qquad \begin{array}{r} 1 \\ +\,1 \\ \hline 10 \end{array}$$

Example

signed int var = -36;

Step 1: Binary equivalent of 36 is $(100100)_2$

Step 2: **1's complement** - inverting binary bits (i.e. switching 0 to 1 and 1 to 0)

1's complement of 36

Step 3: **2's complement** - just add 1 to 1's complement

MSB = 1 indicates negative number

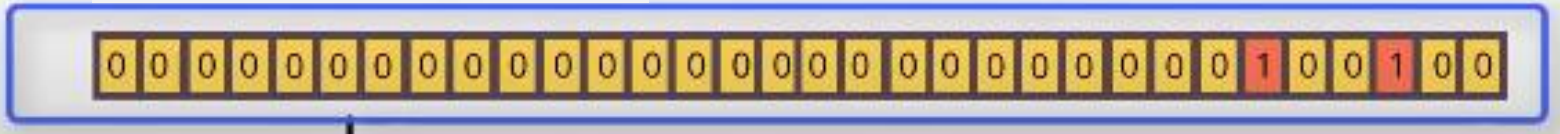
MSB = 1 indicates negative number

2's complement

```c
int main()
{
    signed int var = -36;
    printf("%d", var);

    return 0;
}
```

%d
prints Signed Integer



covert binary to decimal

-36 prints the decimal results by adding - (minus) sign